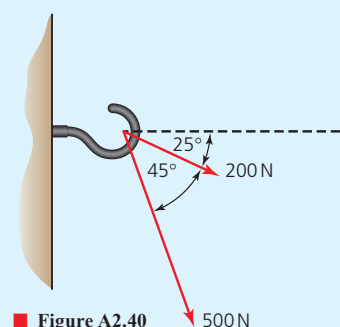


- 28** Parallel forces of 1 N, 2 N and 3 N can act on an object at the same time. State the values of all the possible resultant forces.
- 29** Calculate the resultant force (size and direction) of 4.7 N and 5.9 N which are perpendicular to each other and acting away from a point mass.
- 30** Show that a mass on an inclined plane will just begin to slip down the slope when the tangent of the angle to the horizontal equals the coefficient of static friction.

- 31** Determine by scale drawing or calculation the size and direction of the resultant force acting on the hook shown in Figure A2.40.



■ Figure A2.40

◆ **Newton's laws of motion** **First law:** an object will remain at rest, or continue to move in a straight line at a constant speed, unless a resultant force acts on it; **Second law:** acceleration is proportional to resultant force; **Third law:** whenever one body exerts a force on another body, the second body exerts exactly same force on the first body, but in the opposite direction.

◆ **Balanced forces** If an object is in mechanical equilibrium, we describe the forces acting on it as 'balanced'.

◆ **Equilibrium** An object is in equilibrium if it is unchanging under the action of two or more influences (e.g. forces). Different types of equilibrium include **translational**, **rotational** and **thermal**.

◆ **Translational** Changing position.



■ Figure A2.41 The object is in translational equilibrium, but not in rotational equilibrium

Newton's laws of motion

SYLLABUS CONTENT

- Newton's three laws of motion.

Newton's three laws of motion are among the most famous in classical physics. They describe the relationships between force and motion. Although they were first stated more than three hundred years ago, they are equally important today and are essential for an understanding of all motion (except when a speed of motion is close to the speed of light, as discussed in Topic A.5).

Newton's first law of motion

Newton's first law of motion states that an object will remain at rest or continue to move in a straight line at a constant speed, unless a resultant force acts on it.

In other words, a resultant force will produce an acceleration (change in velocity).

When the influences on any system are **balanced**, so that the system does not change, we describe it as being in **equilibrium**. (As another example, if an object stays at the same temperature, we say that it is in **thermal equilibrium**.)

When there is no resultant force on an object, we say that it is in **translational equilibrium**.

The term **translational** refers to movement from place to place. An object is in translational equilibrium if it remains at rest or continues to move with a constant velocity (in a straight line at a constant speed), as described by Newton's first law.

In passing, it should be noted that, if equal forces act in opposite directions, an object will be in translational equilibrium, but if the forces are not aligned (see Figure A2.41) then the object may start to rotate, so it will not be in **rotational equilibrium**. The subject of rotational dynamics is covered in Topic A.4.

Nature of science: Observations



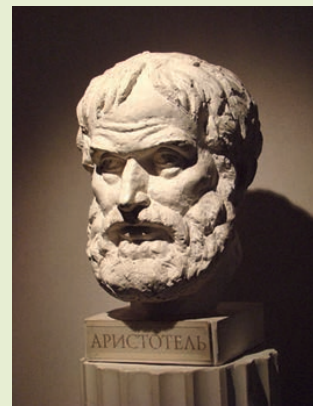
Natural philosophy

'Forces are needed to keep an object moving and, without those forces, movement will stop.' This accepted 'fact' is not true, but it is still widely believed. It was the basis of theories of motion from the time of Aristotle (about 2350 years ago) until the seventeenth century, when scientists began to understand that the forces of friction were responsible for stopping movement.

Aristotle is one of the most respected figures in the early development of human thought. He appreciated the need for wide-ranging explanations of natural phenomena but the 'science' of that time – called natural philosophy – did not involve careful observations, measurements, mathematics or experiments.

Aristotle believed that everything in the world was made of a combination of the four elements of earth, fire, air and water. The Earth was the centre of everything and each of the four Earthly elements had its natural place. When something was not in its natural place, then it would tend to return – in this way he explained why rain falls, and why flames and bubbles rise, for example.

Modern science (characterized by experimentation and the development of unbiased, testable theories) began in the seventeenth century. It includes the work of famous physicists mentioned in this topic: Hooke, Galileo and Newton.



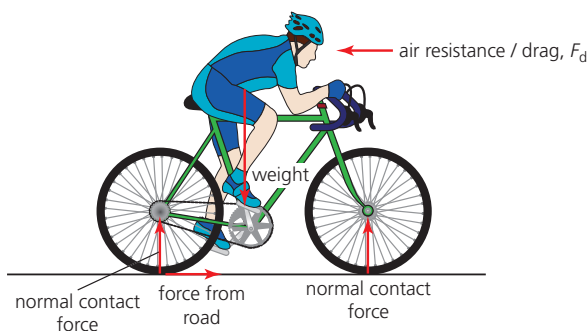
■ **Figure A2.42** A representation of Aristotle

◆ Natural philosophy

The name used to describe the (philosophical) study of nature and the universe before modern science.

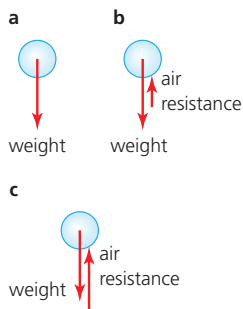
Examples of translational equilibrium

Because all objects on Earth have weight, it is not possible for an object to be in equilibrium because there are no forces acting on it. So, all translational equilibrium arises when two or more forces are balanced.



■ **Figure A2.43** A cyclist moving at constant speed in translational equilibrium

- A book on a horizontal table (Figure A2.8) is in equilibrium because its downwards weight is balanced by the upwards normal contact force.
- A stationary block on a slope (Figures A2.38 and A2.39) is in equilibrium because the component of its weight down the slope is balanced by surface friction up the slope and the component of its weight into the slope is balanced by the normal component of the contact force.
- A cyclist moving with constant speed (Figure A2.43) is in equilibrium because their weight is balanced by the sum of the two normal contact forces and the frictional force from the road is balanced by the drag.



■ **Figure A2.44** The resultant force on a falling object changes as it gains speed

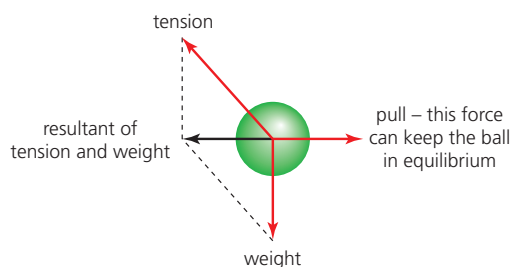
Falling through the air at terminal speed

Figure A2.44 shows three positions of a falling ball. In part **a** the ball is just starting to move and there is no air resistance / drag. In part **b** the ball has accelerated and has some air resistance acting against its motion, but there is still a resultant force and an acceleration downwards. In part **c** the speed of the falling ball has increased to the point where the increasing air resistance has become equal and opposite to the weight. There is then no resultant force and the ball is in translational equilibrium, falling with a constant velocity called its terminal velocity or terminal speed. (Any buoyancy forces are considered to be negligible under these circumstances.) Terminal speed was introduced in Topic A.1.

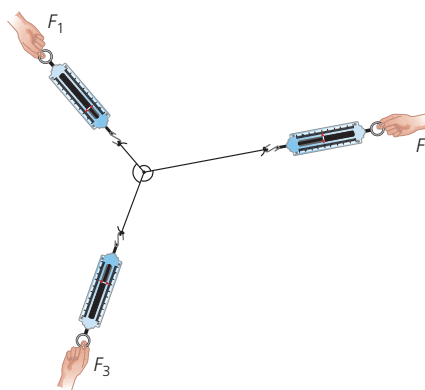
Three forces in equilibrium

If two forces are acting on an object such that it is not in equilibrium, then to produce equilibrium a third force can be added that is equal in size to the resultant of the other two, but in the opposite direction. All three forces must act through the same point. For example, Figure A2.45 shows a free-body diagram of a ball on the end of a piece of string kept in equilibrium by a sideways pull that is equal in magnitude to the resultant of the weight and the tension in the string.

The translational equilibrium of three forces can be investigated in the laboratory simply by connecting three force meters together with string just above a horizontal surface, as shown in Figure A2.46. The three forces and the angles between them can be measured for a wide variety of different values, each of which maintains the system stationary.



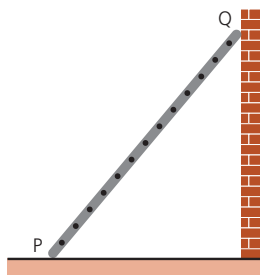
■ **Figure A2.45** Three forces keeping a suspended ball in equilibrium



■ **Figure A2.46** Investigating three forces in equilibrium

WORKED EXAMPLE A2.7

A ladder is leaning against a wall, as shown in Figure A2.47. Friction at point P is stopping the ladder from slipping, but there is no need for any friction acting at point Q.

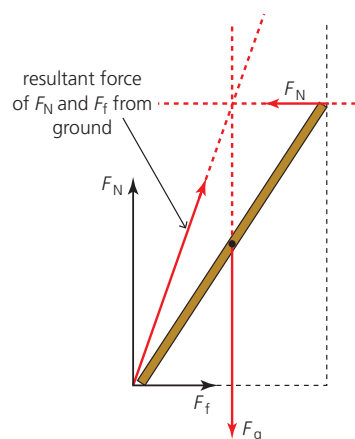


■ **Figure A2.47** A ladder leaning against a wall

- Draw a free-body diagram of the ladder, including its weight and the normal force from the wall.
- The resultant force on the ladder from the ground must be directed at the point where the lines of action of the other two forces intersect. Add this line to your diagram.

- c Complete the diagram by adding the two perpendicular components of the force from the ground on the ladder.

Answer



■ **Figure A2.48**

- 32 Under what circumstances will a moving car be in translational equilibrium?
- 33 If you are in an elevator (lift) without windows discuss whether it is possible to know if you are moving up, moving down or stationary.
- 34 Figure A2.49 shows a mountain climber who, at that moment, is stationary.
- Draw a free-body diagram that shows that he is in equilibrium.
 - Outline the features of your diagram which show that the climber is in equilibrium.
- 35 Can the Moon be described as being in translational equilibrium? Explain your answer.



■ Figure A2.49

■ Newton's second law of motion

We have seen that Newton's first law establishes that there is a connection between resultant force and acceleration. Newton's second law takes this further and states the mathematical connection: when a resultant force acts on a (constant) mass, the acceleration is proportional to the resultant force: $a \propto F$.

Both force and acceleration are vector quantities and the acceleration is in the same direction as the force.

Investigating the effects of different forces and different masses on the accelerations that they produce is an important part of most physics courses, although reducing the effects of friction is essential for consistent results.

Inquiry 1: Exploring and designing

Exploring

Aristotle's understanding of motion was formed through making observations of the behaviour of objects in motion, but without any deep understanding of the concept of force he was unable to account for the effects of friction or air resistance. What methods are available for reducing friction in investigations into the effects of different forces and masses on an object's acceleration?

In groups, brainstorm how experiments can be designed to reduce or to cancel the effects of frictional forces. Decide on a selection of search terms or phrases that can be used by individual students for internet research. Use your research to formulate a research question and hypothesis.

Such experiments also show that when the same resultant force is applied to different masses, the acceleration produced is inversely proportional to the mass, m : $a \propto 1/m$

Combining these results, we see that acceleration, $a \propto \frac{F}{m}$.

Newton's second law can be written as: $F \propto ma$

If we define the SI unit of force, the newton, to be the force that accelerates 1 kg by 1 m s^{-2} , then we can write: force (N) = mass (kg) $\times$ acceleration (m s^{-2})

Newton's second law of motion: resultant force, $F = ma$

◆ **Proportional relationship** Two variables are (directly) proportional to each other if they always have the same ratio.

◆ **Uncertainty bars** Vertical and horizontal lines drawn through data points on a graph to represent the uncertainties in the two values.

This version of Newton's second law assumes that the mass of the object is constant. We will see later in this topic that there is an alternative version which allows for changing mass.

When discussing a gravitational force, weight, we have used the symbol F_g and the acceleration involved is g , the acceleration of free fall.

So, the equation $F = ma$ becomes the familiar:

$$F_g = mg$$

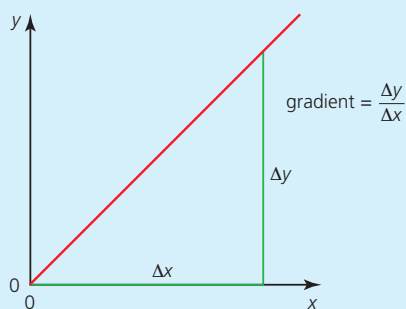
Tool 3: Mathematics

On a best-fit linear graph, construct lines of maximum and minimum gradients with relative accuracy (by eye) considering all uncertainty bars

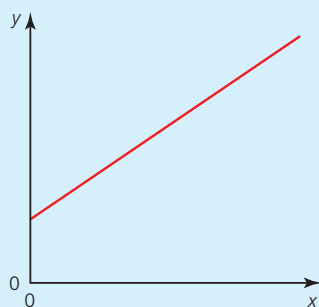
Many basic physics experiments are aimed at investigating if there is a **proportional relationship** between two variables, and this is usually best checked by drawing a graph.

If two variables are (directly) proportional, then their graph will be a straight line passing through the origin

Figure A2.50 represents a proportional relationship. It is important to stress that a linear graph that does not pass through the origin does *not* represent proportionality (Figure A2.51).



■ **Figure A2.50** A proportional relationship

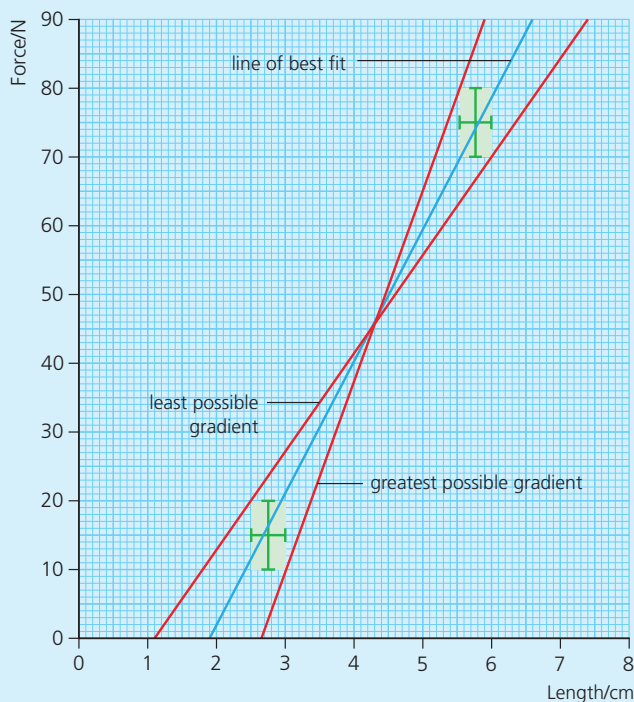


■ **Figure A2.51** A linear relationship that is not proportional does not pass through the origin. See also Tool 3: Mathematics (Understand direct and inverse proportionality) on page 129.

Uncertainty in gradients and intercepts

It is often possible to draw a range of different straight lines, all of which pass through the **uncertainty bars** representing experimental data.

We usually assume that the line of best fit is midway between the lines of maximum possible gradient and minimum possible gradient. Figure A2.52 shows an example (for simplicity, only the first and last error bars are shown, but in practice all the error bars need to be considered when drawing the lines).



■ **Figure A2.52** Finding maximum and minimum gradients for a spring-stretching experiment

Figure A2.52 shows how the length of a metal spring changed as the force applied was increased. We know that the measurements were not very precise because the uncertainty bars are large. The line of best fit has been drawn midway between the other two. This is a linear graph (a straight line) and it is known that the gradient of the graph represents the force constant (stiffness) of the spring and the horizontal intercept represents the original length of the spring. Taking measurements from the line of best fit, we can make the following calculations:

$$\text{force constant} = \text{gradient} = \frac{(90 - 0)}{(6.6 - 1.9)} = 19 \text{ Ncm}^{-1}$$

$$\text{original length} = \text{horizontal intercept} = 1.9 \text{ cm}$$

To determine the uncertainty in the calculations of gradient and intercept, we need only consider the range of straight lines that could be drawn through the first and last error bars. The uncertainty will be the maximum difference between these extreme values obtained from graphs of maximum and minimum possible gradients and the value calculated from the line of best fit. In this example it can

be shown that: force constant is between 14 Ncm^{-1} and 28 Ncm^{-1} , original length is between 1.1 cm and 2.6 cm .

The final result can be quoted as:

force constant = $19 \pm 9 \text{ Ncm}^{-1}$, original length = $1.9 \pm 0.8 \text{ cm}$. Clearly, the large uncertainties in these results confirm that the experiment lacked precision.

Table A2.3 shows the results that a student obtained when investigating the effects of a resultant force on a constant mass. Plot a graph of these readings, including uncertainty bars. Then draw lines of maximum and minimum gradients through the error bars. Finally, use your graph to determine the mass that the student used in the experiment and the uncertainty in your answer.

■ Table A2.3

Resultant force, N, $\pm 0.5 \text{ N}$	Acceleration, m s^{-2} , $\pm 0.2 \text{ m s}^{-2}$
1.0	0.7
2.0	1.3
3.0	2.0
4.0	2.8
5.0	3.3
6.0	4.1

Common mistake

Many students believe that the force involved when an object hits the ground is its weight. In reality, the force will depend on the nature of the impact. The longer the duration of the impact, the smaller the force, as explained below.

Non-mathematical applications of Newton's second law

We can use Newton's second law to explain why, for example, a glass will break when dropped on the floor, but may survive being dropped onto a sofa. A collision with the floor will be for a much shorter duration, which means the deceleration will be greater and (using $F = ma$) the force will be greater, and probably more destructive. Similar arguments can be used to explain how forces can be reduced in road accidents.

WORKED EXAMPLE A2.8

A car of mass 1450 kg is accelerated from rest by an initial resultant force of 3800 N .

- Calculate the acceleration of the car.
- If the force and acceleration are constant, what will its speed be after 4.0 s ?
- Determine how far it will have travelled in this time.
- After 4.0 s the resistive forces acting on the car are 1800 N . Show that the new force required to maintain the same acceleration is approximately 5.5 kN .

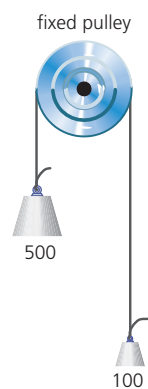
Answer

- $a = \frac{F}{m} = \frac{3800}{1450} = 2.62 \text{ m s}^{-2}$
- $v = u + at = 0 + (2.62 \times 4.0) = 10.5 \text{ m s}^{-1}$
- $s = \frac{(u + v)}{2} \times t = \frac{(0 + 10.5)}{2} \times 4.0 = 21.0 \text{ m}$
- $3800 + 1800 = 5600 \text{ N} \approx 5500 \text{ N} = 5.5 \text{ kN}$

WORKED EXAMPLE A2.9

Figure A2.53 shows two masses attached by a string which passes over a fixed pulley. Assuming that there is no friction in the system and that the string has negligible mass, determine:

- a the acceleration of the system
- b the tension in the string.



■ **Figure A2.53** Two masses attached by a string which passes over a fixed pulley

Answer

- a The resultant force on the system of two masses = weight of the 500 g mass – weight of 100 g mass = $(0.500 - 0.100) \times 9.8 = 3.9 \text{ N}$

$$a = \frac{F}{m} = \frac{3.9}{(0.500 + 0.100)} = 6.5 \text{ ms}^{-2}$$

The 500 g mass will accelerate down while the 100 g mass accelerates up at the same rate.

- b Consider the 100 g mass: the resultant force acting = tension, T , in the string upwards – weight acting downwards = $T - (0.100 \times 9.8) = T - 0.98$

$$F = ma$$

$$(T - 0.98) = 0.100 \times 6.5$$

$$T = 1.6 \text{ N}$$

Equally, we could consider the 500 g mass: the resultant force acting = weight acting downwards – tension, T , in the string upwards =

$$(0.500 \times 9.8) - T = 4.9 - T$$

$$F = ma$$

$$(4.9 - T) = 0.500 \times 6.5$$

$$T = 1.6 \text{ N}$$

- 36 A laboratory trolley accelerated at 80 cm s^{-2} when a resultant force of 1.7 N was applied to it. What was its mass?
- 37 When a force of 6.4 N was applied to a mass of 2.1 kg on a horizontal surface, it accelerated by 1.9 ms^{-2} . Determine the average frictional force acting on the mass.
- 38 When a hollow rubber ball of mass 120 g was dropped on a concrete floor the velocity of impact was 8.0 ms^{-1} and it reduced to zero in 0.44 s (before bouncing back).
 - a Calculate:
 - i the ball's average deceleration
 - ii the average force exerted on the ball.
 - b Repeat the calculations for a solid steel ball of the same size, 10 times the mass, but with the same impact velocity. Assume that its speed reduced to zero in 0.080 s .
 - c Outline why the steel ball can do more damage to a floor than the rubber ball.

- 39 A small aircraft of mass 520 kg needs to take off with a speed of 30 ms^{-1} from a runway in a distance of 200 m .

- a Show that the aircraft needs to have an average acceleration of 2.3 ms^{-2} .
- b What average resultant force is needed during the take off?

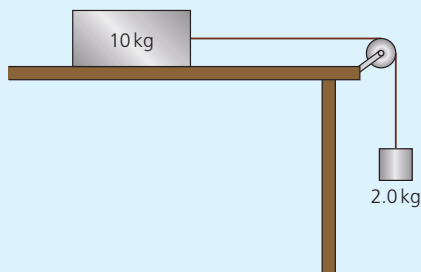
- 40 Discuss why the forces on the long-jumper shown in Figure A2.54 are reduced because he is landing in sand.



■ **Figure A2.54** Impact in a sand-pit reduces force

- 41 a What resultant force is needed to accelerate a train of total mass $2.78 \times 10^6 \text{ kg}$ from rest to 20 m s^{-1} in 60 s?
- b If the same train was on a sloping track which had an angle of 5.0° to the horizontal, what is the component of its weight parallel to the track?
- c Suggest why railway designers try to avoid hills.
- 42 Calculate the average force needed to bring a 2160 kg car travelling at 21 m s^{-1} to rest in 68 m.
- 43 Use Newton's second law to explain why it will hurt you more if you are struck by a hard ball than by a soft ball of the same mass and speed.
- 44 A trolley containing sand is pulled across a frictionless horizontal surface with a small but constant resultant force. Describe and explain the motion of the trolley if sand can fall through a hole in the bottom of the trolley.
- 45 A man of mass 82.5 kg is standing still in an elevator that is accelerating upwards at 1.50 m s^{-2} .
- a What is the resultant force acting on the man?
- b What is the normal contact force acting upwards on him from the floor?

- 46 Figure A2.55 shows two masses connected by a light string passing over a pulley.
- a Assuming there is no friction, calculate the acceleration of the two blocks.
- b What resultant force is needed to accelerate the 2.0 kg mass by this amount?
- c Draw a fully labelled free-body diagram for the 2 kg mass, showing the size and direction of all forces.



■ **Figure A2.55** Two masses connected by a light string passing over a pulley

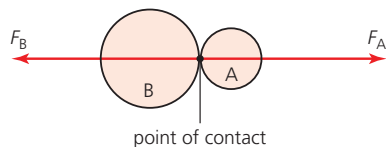
- 47 Outline how air bags (and/or seat belts) reduce the injuries to drivers and passengers in car accidents.

◆ **Inertia** Resistance to a change of motion. Depends on the mass of the object.

Newton's second law offers us a different way of understanding mass: larger masses accelerate less than smaller masses under the action of the same resultant force. So, mass can be considered as a measure of an object's resistance to acceleration. Physicists use the term **inertia** to describe an object's resistance to a change of motion.

Mass is a measure of inertia.

■ Newton's third law of motion



■ **Figure A2.56** When two bodies interact, $F_A = -F_B$

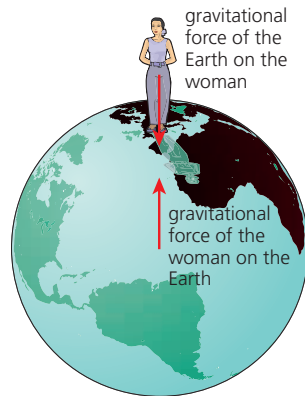
Whenever any two objects come in contact with each other, or otherwise interact, they exert forces on each other (Figure A2.56). Newton's third law compares these two forces.

Newton's third law of motion states that whenever one body exerts a force on another body, the second body exerts a force of the same magnitude on the first body, but in the opposite direction.

Essentially this law means that forces must always occur in equal pairs, although it is important to realize that the two forces must act on different bodies and in opposite directions, so that only one of each force pair can be seen in any free-body diagram. The two forces are always of the same type, for example gravity/gravity or friction/friction. Sometimes the law is quoted in the form used by Newton: 'to every action there is an equal and opposite reaction'. In everyday terms, it is simply not possible to push something that does not push back on you. Here are some examples:

- If you pull a rope, the rope pulls you.
- If the Earth pulls a person, the person pulls the Earth (Figure A2.57).
- If a fist hits a cheek, the cheek hits the fist (Figure A2.58).

- If you push on the ground, the ground pushes on you.
- If a boat pushes down on the water, the water pushes up on the boat.
- If the Sun attracts the Earth, the Earth attracts the Sun.
- If an aircraft pushes down on the air, the air pushes up on the aircraft.



■ **Figure A2.57** The force on the woman is equal and opposite to the force on the Earth



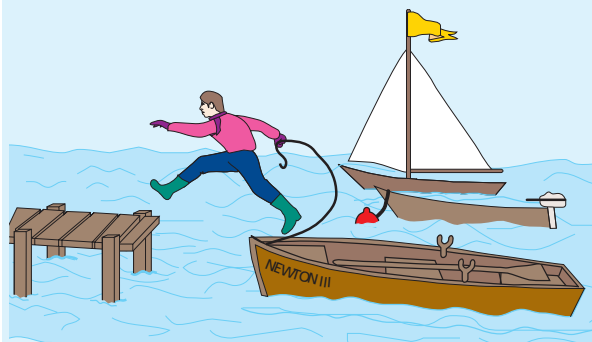
■ **Figure A2.58** The force on the glove is equal and opposite to the force on the cheek

48 A book has a weight of 2 N and is at rest on a table. The table exerts a normal contact force of 2 N upwards on the book.
Explain why these two forces are *not* an example of Newton's third law.

49 Seven examples of pairs of Newton's third law forces are provided above. Give three more examples. Try to use different types of force.

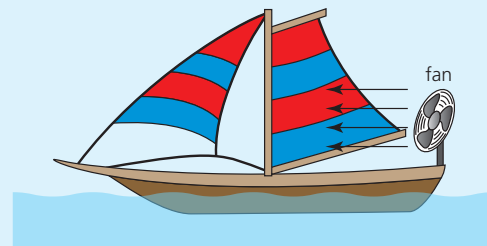
50 Consider Figure A2.58. Outline reasons why forces of equal magnitude, for example on a face and on a fist, can have very different effects.

51 Discuss why the person shown in Figure A2.59 could end up in the water.



■ **Figure A2.59**

52 Figure A2.60 shows a suggestion to make a sailing boat move when there is no wind.
Discuss how effective this method could be.



■ **Figure A2.60** Sailing boat

53 A large cage with a small bird sitting on a perch is placed on weighing scales.
Discuss what happens to the weight shown on the scales when the bird is flying around the cage (compared with when it is sitting still).